



SC-RRT: Bidirectional Adaptive Superellipsoid Constrained Sampling for Path Planning in Cluttered Obstacle Environments

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Abstract

针对传统快速探索随机树 (RRT) 算法中多节点仅用于环境探索且缺乏路径优化导向的问题, 本文提出了一种自适应双向超椭球体约束采样 (SC-RRT) 算法, 以在复杂障碍物环境中实现高效、可行的路径规划, 并平衡收敛时间与路径长度这两大优化目标。

SC-RRT算法首先根据起点与目标点的环境位置信息自适应选择中介点, 构建“起点-中介点-目标点”的双段路径框架, 并通过双向并行扩展策略 (起点→中介点、目标点→中介点) 提升搜索效率。然后, 算法对两段路径进行长度预估与优化, 将优化后的路径长度作为启发式子集的长轴直径, 约束新节点在该子集内采样, 从而实现路径的定向优化。此外, SC-RRT引入自适应采样范围控制机制, 通过三维度的帕累托支配准则动态筛选优势节点, 并将其更新为新的起点, 从而将全局规划问题转化为分阶段子问题, 进一步提升路径优化效果。仿真结果表明, SC-RRT算法在收敛时间与路径长度之间取得了优异的平衡。与传统RRT算法相比, 其收敛速度显著提高; 与RRT-Connect算法相比, 路径质量和搜索鲁棒性表现更优; 与启发式RRT*算法相比, SC-RRT在生成相同长度最优路径时, 计算时间减少超过30%, 验证了其在复杂环境中的优越性能和工程实用性。

Keywords Rapidly exploring Random Trees (RRT) · Informed-RRT · RRT-connect · Pareto Dominance

1 Introduction

随着工业4.0与服务机器人技术的迅猛发展, 智能机器人已成为提升制造、服务等领域自动化水平与作业效率的核心载体, 其应用场景已从结构化工厂延伸至动态复杂的人机协作环境[1–3]。这一趋势对机器人的自主决策能力提出了更高要求, 尤其是作为核心技术模块的路径规划算法, 其性能直接决定机器人作业的安全性、可靠性与实时性[4–6]。因此, 针对复杂环境下的机器人路径规划算法展开深入研究, 不仅具有重要的理论价值, 更对推动智能机器人的工程化应用具有关键意义[7, 8]。路径规划的核心任务是: 在已知起点与目标点的配置空间中, 为机器人生成一条满足运动约束的无碰撞轨迹, 同时平衡“实时响应”与“路径最优”两大核心目标[9]。这一任务在动态场景下更凸显其挑战性: 一方面, 环境中随机分布的动态障碍物与机械臂自身运动的不确定性, 要求算法具备快速的避障决策能力; 另一方面, 工业机械臂的高精度作业与服务机器人的人机安全交互需求, 又对路径的平滑性与最优性提出了严苛标准[10–12]。当前, 如何在动态障碍物分布环境中, 同时实现“快速收敛”与“高质量路径”的协同优化, 已成为路径规划领域亟待突破的核心瓶颈。

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为应对上述挑战，国内外学者围绕快速探索随机树（Rapidly exploring Random Trees, RRT）这一经典采样类算法，提出了一系列改进策略，形成了“最优性优化”与“收敛加速”两大研究分支。在路径最优性探索方面，Karaman等提出的RRT*算法具有里程碑意义——该算法通过引入父节点重选与树重布线（rewiring）机制，打破了传统RRT“无最优性保证”的局限，实现了路径的渐进最优性[13]。然而，这种最优性是以牺牲收敛速度为代价的，树结构的频繁调整导致其在动态环境中难以满足实时性需求[14]。

针对RRT*收敛缓慢的问题，“知情采样”（Informed Sampling）策略为算法加速提供了新的思路，其核心思想是：在获得初始可行路径后，将后续采样范围收缩至可能包含更优解的启发式区域内，从而减少无效采样[15]。Gammell等提出的Informed RRT*算法是该方向的代表性成果，其创新性地引入椭圆启发域（Ellipsoidal Informed Region）概念，将采样空间约束在以起点、目标点为焦点，以初始路径长度为长轴的椭圆内，显著提升了收敛效率[16]。但该策略的性能深度依赖初始路径的质量：若初始路径偏离全局最优解过多，椭圆启发域会将搜索局限于局部区域，导致算法陷入局部最优困境，难以生成真正的最优路径[17–20]。

为突破初始路径质量的制约，学者们转向“快速生成高质量初始路径”的研究，形成了目标偏置与势场混合两大技术路线。目标偏置策略通过提高向目标区域采样的概率，为随机树生长提供明确导向，有效缩短了首次找到可行路径的时间[21–22]。更具代表性的是人工势场法与RRT的混合策略：该方法利用势场对目标的“引力”与对障碍物的“斥力”构建引导场，将随机采样转化为有向搜索，不仅能快速生成初始路径，还能构建高效的采样集合。已有研究证实，该混合策略的收敛速度较Informed RRT*平均提升约62.5%[23]，但路径质量仍存在波动，且在障碍物密集区域易出现搜索方向震荡的问题[24]。

近年来，尽管LBT-RRT（Lower Bound Tree-RRT）等改进算法通过分层搜索机制，实现了“收敛速度优于RRT*、路径质量优于基础RRT”的突破[28]，但在随机障碍物分布的复杂环境中，其仍存在两大局限：一是路径优化过程与采样引导的协同性不足，导致收敛后期效率下降；二是对动态环境的适应性较弱，难以平衡“快速响应”与“路径最优”的矛盾[29]。

针对现有算法的上述缺陷，本文提出一种基于自适应双向超椭球体约束采样的路径规划算法（Superellipsoid-Constrained Bidirectional RRT, SC-RRT）。该算法通过创新的采样约束机制自适应地收缩采样范围与双向搜索策略，将启发式路径长度优化与双向树扩展结合，从而有效提高新节点的生成效率并加速收敛过程。在保证路径长度优化的同时，显著提高了收敛速度，特别适用于动态障碍物和复杂环境下的路径规划问题。此外，SC-RRT还引入了基于动态规划的思想，通过帕累托支配准则对节点进行筛选，优化了路径生成的速度和精度。本文的主要贡献在于提出了SC-RRT算法，该算法不仅提高了收敛速度，还优化了路径质量，适用于处理随机障碍物分布的复杂场景。具体来说：

(1)采用了类似Informed-RRT*算法中的启发式子集概念，但不同的是，SC-RRT通过动态估算路径长度，并基于此对采样范围进行自适应限制，从而优化了路径规划的效率。

(2)引入了动态规划和帕累托支配准则，使得SC-RRT能够在多个局部最优解中选择最优节点，从而提高了路径生成的效率并优化了收敛速度。

(3)SC-RRT在保证路径长度的前提下，能够在动态障碍物和复杂环境中提供更为稳定且快速的规划结果。

本文的余下部分结构安排如下：第2节阐述了路径规划问题的定义并介绍了相关术语；第3节详细说明了针对SC-RRT算法的具体改进措施，并给出了相应的伪代码；第4节通过MATLAB仿真实验，评估了SC-RRT算法在平衡收敛速度与路径长度方面的性能表现；最后，第5节总结了全文内容并指出了未来的研究方向。

2 Problem statement and notation

2.1 Problem statement

通常，路径规划问题可分为静态环境下的路径规划与动态环境下的路径规划。静态路径规划指在已知环境模型中，寻找一条从起点到目标点的连续且无碰撞的路径；而动态路径规划则是在初始路径生成的基

基础上，借助额外程序或传感器实时感知环境，动态规避检测到的障碍物。需要指出的是，无论是静态还是动态路径规划，均需首先完成全局路径的静态规划。因此，静态路径规划在路径规划研究中具有不可替代的基础性地位。尽管动态规划在算法实现、硬件要求和功能复杂性方面通常更具挑战，静态路径规划仍被普遍视为该领域的核心问题。有鉴于此，本文聚焦于静态环境下的路径规划研究，下文中所讨论的路径规划问题，如无特别说明，均指静态路径规划。

为构建路径规划问题的数学模型，本文定义以下两个核心空间：工作空间 $W \subseteq \mathbb{R}^m$ 与构型空间 $C \subseteq \mathbb{R}^n$ ，其中 m 与 n 分别表示两空间的维度。 W 为机器人实际运动的笛卡尔空间，而 C 则描述机器人所有可能的构型。两者在环境表示上具有等价性，均完整表达了自由区域、障碍物及环境边界信息。为便于描述，下文将主要基于工作空间展开讨论，但所有论述同样适用于构型空间。

定义 $x \in W$ 为工作空间中的一个状态。将整个空间划分为无碰撞区域 W_{free} 与障碍区域 W_{obs} ，满足 $W = W_{\text{free}} \cup W_{\text{obs}}$ 。路径规划的起点与终点与中介点分别记为 x_s, x_g, x_{mid} ，且 $x_s, x_g, x_{\text{mid}} \in W_{\text{free}}$ ，令 Π 表示所有路径的集合， Π_{free} 则表示所有无碰撞路径的集合。一条路径可定义为

$$\delta(t) = \{\delta(a) = x_{\text{start}}, \delta(b) = x_{\text{mid}}, \delta(c) = x_{\text{goal}}, \forall t \in [a, c], \delta(t) \in \Pi_{\text{free}}\} \quad (\text{式1})$$

式中 $\delta \in \Pi_{\text{free}}$ 表示生成的可行路径，且任意 $t \in [a, c]$ 均满足 $\delta(t) \in \Pi_{\text{free}}$ ，在上述数学模型的基础上，路径规划的核心目标是在有限的时间内于复杂的工作空间 W 中构建一棵从起点 x_s 扩展到目标区域 x_g 的随机树，以寻找一条满足无碰撞约束的可行路径。针对这一问题，采样式规划算法因其对高维空间的良好适应性与对障碍环境的鲁棒性，成为目前最具代表性的解决途径之一。

既往研究中已严格证明RRT具有概率完备性（probabilistic completeness）：对于任意的路径规划问题，若存在一个可行解，当迭代次数 N 趋于无穷时，算法能够以概率 1 找到一个满足约束的解。其数学表达式如下：

$$P \left(\left\{ \begin{array}{c} \exists x_{\text{start}}, x_{\text{goal}} \in X_{\text{free}} \\ \cap \\ \lim_{i \rightarrow \infty} (V_i \cap x_{\text{goal}}) \neq \emptyset \end{array} \right\} \right) = 1 \quad (\text{式2})$$

其中， V_i 表示第 i 次迭代后树结构中的节点集合。该结论说明：随着迭代次数不断增加，RRT 终将以概率 1 到达目标区域，从而得到一条可行路径。

而其改进算法RRT*则进一步满足渐进最优性（asymptotic optimality）：对于任意可行规划问题，随着迭代次数 $i \in \mathbb{N}$ 趋于无穷，则算法构建的最优路径代价函数 C_i 将收敛 $\Sigma_{\text{free}} \rightarrow R \geq 0$ 将持续收敛到某个有限的最优值 $C_{\text{best}} \in R \geq 0$ 。即非负实数集合

$$P(\lim_{i \rightarrow \infty, \delta \in \Sigma_{\text{free}}} C_i^\delta = C_{\text{best}}) = 1 \quad (\text{式3})$$

上述性质表明，RRT 以探索速度见长，可在有限迭代次数内以概率 1 捕获可行解，却无法对解的路径长度给出确定性上界，其随机扩展的特性使得大部分采样点往往与最优路径关联不大，从而导致解的质量难以保证；相反，RRT* 通过“重写（rewiring）”机制持续缩短路径代价，当采样次数趋于无穷时，其输出序列几乎必然收敛至全局最优解，但计算开销随之显著增加。这一本质差异促使后续工作聚焦在“如何兼顾 RRT 的快速扩张与 RRT* 的渐近最优”问题上。

近期文献表明，在重要性测度（informed measure）约束下进行定向采样能够显著提升新节点质量并降低无效扩展率，从而加快最优解的收敛速度。为此，构造一个紧凑且充分覆盖最优解空间的知情子集

(informed set) 成为关键前提。典型算法 Informed-RR通过在线构造椭球形知情子集 (informed set) 将采样空间由 X_{free} 压缩至“可能包含最优解”的子空间，从而在保持渐进最优性的同时显著减少无效采样。然而，该策略仍然面临两大瓶颈：其一，知情椭圆依赖已得最优路径长度作为半轴，初期若解质量差，则知情区域依旧庞大，导致采样效率呈“先低后高”的非均匀特性，难以在搜索初期具备高效率；其二，椭圆内部采用均匀采样，未顾及状态空间中的代价梯度与障碍分布，仍可能造成大量节点诞生于高代价或窄通道区域，影响整体扩展效率。

基于上述分析，可以看出现有基于知情采样的 RRT* 改进方法在搜索效率、采样分布质量与环境适应性方面仍存在提升空间。为此，本文提出SC-RRT在 Informed-RRT* 的基础上引入三阶段改进：

(1) 自适应预测纠偏阶段控制椭圆收缩：通过针对代价预估函数提出误差反馈-动态调整的在线学习思想，使采样过程能够实时感知环境复杂度并修正自身偏差。通过比例P、积分I、微分D环节的协同，得到最终修正后的路径总代价预估 $F_{PID}(c)$ ，用于定义超椭球体的长轴，直接避免代价预估不准导致的超椭球体“虚大”或“过小”避免预估偏差导致的无效采样。同时为了防止 $F_{PID}(c)$ 突变导致采样区域突增或骤减，对其做上下界限制 $\{F_{PID}(c) | F(c) \times 0.7 \leq F_{PID}(c) \leq F(c) \times 0.2\}$

$$\hat{F}_{PID}(c) = \hat{F}(c) \times (1 + K_p * \tilde{e} + K_i * \sum \tilde{e}(k) + K_d * \frac{\tilde{e}(t) - \tilde{e}(t-1)}{\Delta t}) \quad (\text{式4})$$

式中， e 为当前预估误差， Σe 为累积误差， $\dot{e}$ 为误差变化量

$$e = |F_{real}(c) - \hat{F}(c)| \quad (\text{式5})$$

$$\dot{e} = e(t) - e(t-1) \quad (\text{式6})$$

为便于计算将误差定义归一化后为：

$$\tilde{e}(t) = \frac{e(t)}{\hat{F}_{real} - \varepsilon_{num}} \quad (\text{式7})$$

$$\frac{\tilde{e}(t) - \tilde{e}(t-1)}{\Delta t} \quad (\text{式8})$$

通过将实际与估计的绝对误差转化为无量纲及数值波动更稳定的相对误差，消除随 F 大小剧烈抖动。

(2) 帕累托支配最优节点的选择策略优化：通过对帕累托最优支配节点的选择策略进行优化，引入改进后的路径预估代价 $F_{PID}(c)$ 以及规整化比例因子作为第二第三维度帕累托向量，直接量化环境复杂度，通过环境自适应权重调整实现“探索效率-路径代价-环境适应性”多目标平衡，使其在复杂场景中更精准的筛选最优节点。

$$v = [-o(x), F_{PID}(c), \lambda(x)] \quad (\text{式9})$$

式中， $o(x)$ 为探索代价，负号表示趋向探索稀疏区域， $F_{PID}(c)$ 是预估路径总代价的预测值， $\lambda(x)$ 是规整化比例因子，反映障碍密度与可行性，当规整化比例因子 $\lambda(x) > 1$ 时，表示障碍物密集，路径需绕行。且

$\lambda(x) = \frac{A(x)}{S(x)}$ ， $A(x)$ 是起点到 x 点的实际距离， $S(x)$ 是起点到 x 点的欧式距离。通过帕累托最优策略对扩展节点

进行决策，从而实现探索效率-路径代价-环境适应性的“三目标协同”，使算法在复杂场景中更精确地筛选最佳扩展方向，实现探索与利用的平衡。

(3) 双向超椭球体约束采样增加采样效率：通过采用rrt-connect路径规划算法中双向规划的策略将其应用在知情子集约束采样时双向同时进行规划的过程中，该实现不仅保证高效搜索的范围约束也做到了快

速进行全局路径规划的效率提升。

综上，SC-RRT算法在保留 Informed-RRT* 渐近最优性的同时，引入了多级自适应机制，使其在复杂环境下表现出更优的收敛特性与路径质量。首先，通过在线预测修正模型对代价误差进行动态补偿，使得节点扩展能够实时适配环境变化并抑制过度估计或低估所带来的路径偏差问题，从而提高代价评估的可信度。其次，利用帕累托支配准则对扩展节点实施多目标协同优化，使算法在探索效率、路径代价与环境规整性之间实现平衡，显著提升在障碍密集场景下的扩展精度。再次，通过引入双向树自适应约束策略，进一步提高搜索的连通效率，避免双树在高密度区域过度细化，从而保持高效且稳定的扩展速度。

2.2 Notation

除了上述定义外，还通过以下术语来介绍本文提出的算法和改进之处。

注：记得修改，对齐加入PID控制的优化1、路径代价预估函数HeurCost(Heuristic Cost)：通过给定的局部RRT扩展的节点树中起点与终点，此处作用为改进RRT算法中的局部路径代价预估函数，便于在全局路径规划中复用于双向路径规划的操作。其中此时的起点为 x_{start} 、 $x_{mid} \in W_{free}$ ，此时通过给定条件选取的中介点 x_{mid} 作为局部路径规划时的临时终点进行规划， x_{mid} 每隔 T_{mid} 次迭代重新选择，为避免频繁波动，设定默认值 $T_{mid} = 10$ ，其范围内的扩展节点 $x_{current} \in W_{free}$ ，以及路径代价预估函数 $HeurCost(Heuristic Cost)(x_{start}, x_{mid}, x_{current}, C_{current})$ 返回从起点 x_{start} 到中介点 x_{mid} 所得到的预估路径代价长度 $C_{current} \in \mathbb{R} \geq 0$ 。其被运用来作为超椭球体采样约束的主要长轴范围，从而取限制知情子集的采样约束。

2、超椭球体采样函数Sample Ellipsoid: 为克服原始代价估计偏差导致的超椭球尺寸误判问题，本文提出在Informed-rrt基础上引入在线误差反馈以动态修正估计代价，结合rrt-connect的双向策略同时在起点与选定中介点及目标点 x_{start} 、 x_{mid} 、 x_{target} 构成的 $x_{start} - x_{mid}$ 、 $x_{target} - x_{mid}$ 两条路径区间内构建独立的知情超椭球：

(1)、一是以焦点对 (x_{start}, x_{mid}) ，另一个以 (x_{goal}, x_{mid}) 。

(2)、对每个方向分别采用PID修正的长轴（式4）：若以 c_s (由起点侧选取的扩展节点) 和 c_g (由终点侧选取) 估计，则长轴分别为：

$$2a_s = F_{PID}(c_s), 2a_g = F_{PID}(c_g)$$

(3)、各自焦距为：

$$\varepsilon_s = \|x_{start} - x_{mid}\|, \varepsilon_g = \|x_{goal} - x_{mid}\|$$

(4)、对应短轴为：

$$2b_s = \sqrt{(2a_s)^2 - \varepsilon_s^2}, 2b_g = \sqrt{(2a_g)^2 - \varepsilon_g^2},$$

进行双向并行采样，且两段路径的超椭球体采样范围绘制具有严格的对偶性，起点侧从椭球 ε_s 采样，目标侧从椭球 ε_g 采样。若两侧各自生成的节点能在 $free$ 空间中连接，则完成连接。对每一侧，长轴采用PID校正后的估计 F_{PID} ，短轴则基于椭圆参数 a （半长轴）的范围与焦距 ε 推导并按局部障碍占有率进行自适应收缩。绘制出的两个超椭球体采样空间，会从空间内各返回一个随机采样的节点 x_{rand1} 、 $x_{rand2} \in W_{free}$ ，通过调整帕累托最优支配节点的选择策略，从随机生成的 x_{rand} 中按一定条件选择帕累托最优节点作为新的超椭球体

采样节点的焦点通过该节点作为焦点 F_1 ，将选定的中介点 x_{mid} 作为焦点 F_2 ，焦距为两焦点欧氏距离

$\varepsilon = \|x_{start} - x_{mid}\|$ & $\varepsilon = \|x_{goal} - x_{mid}\|$ 再一次进行超椭球体的绘制去约束采样范围，该方法在二维及三维和随机

障碍场中相比Informed-rrt显著降低了无效采样率并加速了连接成功率。**在实验里要呈现 PID 学习曲线**

（误差随迭代的收敛），并把参数敏感性作为附表。为简化表述、避免冗余，下文将以 “ $x_{start} \rightarrow x_{mid}$ ” 方向为例，详细阐述超椭球体的构建原理、参数计算及采样实现流程。“ $x_{goal} \rightarrow x_{mid}$ ” 方向的超椭球体约束采样，可完全参照该逻辑推导，其核心步骤与性能特性保持一致，不再单独赘述。

故在双向规划的框架中，对于任意一个方向（以起点侧为例），知情子集定义为：

$$\zeta_S(x) = \{x \in W_{free} : G_S(x) + H_s(x) \leq F_{PID}, s(c_s)\}$$

其中：

- $G_s(x)$: 从起点 S 到节点 x 的启发式代价或累积路径代价;
- $H_s(x)$: 从 x 到中介点 X_{mid} 的启发式代价;
- $F_{PID,s}(c_s)$: 从起点侧候选节点 c_s 得到的 PID 修正后路径代价估计。

最终双向知情采样区域为两集合并:

$$\zeta(x) = \zeta_s(x) \cup \zeta_G(x)$$

其几何形状对应为两个独立构建的 m 维超椭球体, 如下图

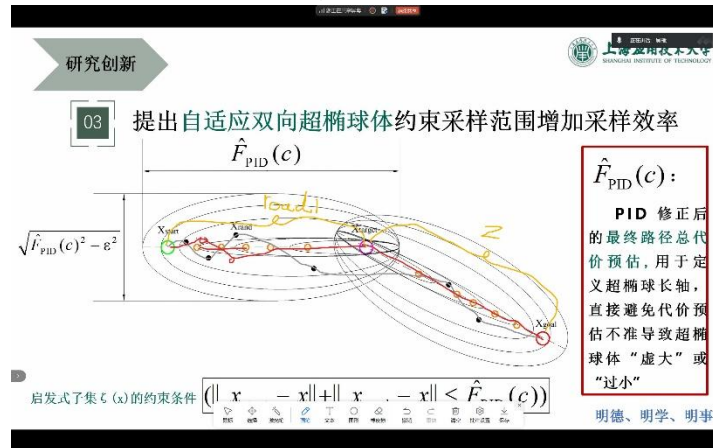


图1 m 维超椭球体

3、Proposed methods

本节主要详细介绍SC-RRT算法, 3.1介绍了通过对价总预估的改进, 用于启发式自适应地限制采样范围。3.2通过对帕累托最优支配节点第三维度环境复杂度判断的优化进行描述, 最后在3.3节中给出SC-RRT算法的伪代码, 并给出简要总结

2 Problem statement and notation

2.1 Problem statement

Generally, path planning problems include static and dynamic path planning. Static path planning can be defined as finding a collision-f

ath length. Even, it could converge to a better path without generating the initial one while Informed RRT* takes more time to compute an equivalent path.

- 1) The stability comparison simulations demonstrate that Dynamic RRT is more stable than the variants mentioned above under various environments and varying distances between the start and goal nodes. In addition, the paths it generates are more concentrated.

3 Conclusion

This paper introduced a Dynamic RRT algorithm for balancing the convergence time and path length in path planning. It improves the new node generation efficiency and quality by constraining to sample directly from an informed subset and decomposing the problem into subproblems. The simulation results show that convergence time and path length can be traded-off by designating the interval of the Pareto dominance procedure, and Dynamic RRT performs better than given several traditional and advanced RRT variants. This means a better path can be generated by this algorithm within the same time or less time as other algorithms. It will save a lot of time for other computing in the application of robots and the strong stability and robustness also make it possible to apply in industrial production.

In future work, an interesting challenge will be to use our method to fit any cost function, as RRT* does, rather than optimizing only the Euclidean distance. It will also be interesting to apply our method to various robots, whether a robot car or a high-degree-of-freedom manipulator, to investigate its superiority in various dimensions.

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Code availability The source code in this paper will be disclosed upon acceptance for publication of this paper. Before accepting, please E-mail to 2,399,496,491@qq.com if the source code is needed.

Declarations

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